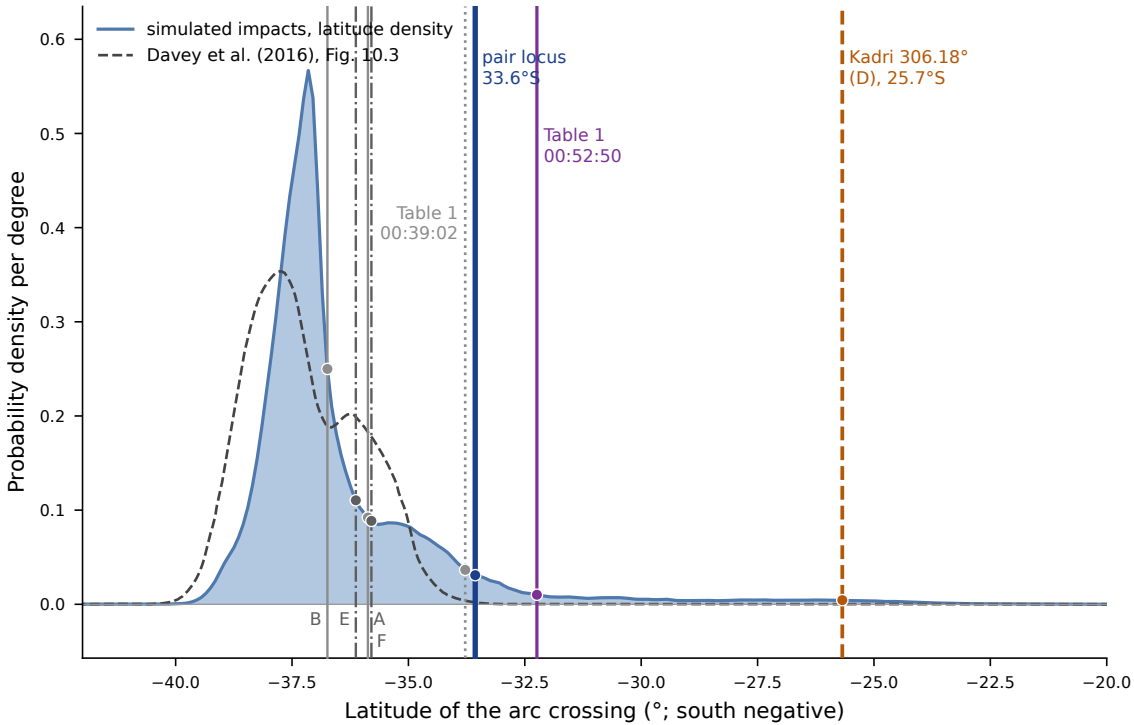

crossing	lat °S	density	±0.5°
G	33.56	0.031	0.033
00:52:50	32.24	0.010	0.010
D	25.68	0.004	0.004
00:39:02	33.78	0.037	0.040
A	35.87	0.092	0.099
B	36.74	0.250	0.315
E	36.13	0.110	0.120
F	35.80	0.089	0.095

A, B: Table 1 bearings 00:49:58, 00:53:31
E: Cape Leeuwin 00:52:04 x Diego Garcia 01:10:01
F: Cape Leeuwin 00:54:27 x Diego Garcia 01:12:49
mass ±0.5°: share of simulated impacts within 0.5° of latitude

(i) The blue curve shows where the simulated impacts are along the arc. The vertical lines show where each sound gives a point on the arc. At the pair point (33.6°S) the curve is about 5 % of its peak height, and 3 % of the simulated impacts are within 0.5° of it. Kadri's direction 306.18° gives 25.7°S, where there are few simulated impacts. The two sounds (Cape Leeuwin 00:52:04, Diego Garcia 01:12:49) have a time difference that agrees with one point on the arc, 33.6°S 94.5°E. Pairs like this occur by chance: the test gives p 0.019 from 36 tests, which is a chance level. The Diego Garcia sound is not kept when the airgun noise is removed. The sound direction that Kadri published for 00:52 does not point to this place. The impact would need about 330 times more sound than the calibration crash gives.

(ii) Pair G: Cape Leeuwin H01W 00:52:04 UTC (DIGITISED Kadri 2024 Fig. 9 panel c; envelope matched filter M1b p 0.043; Kadri's 'text time 00:52:03', 'caption time 00:54:30') x Diego Garcia South H08S 01:12:49 (DIGITISED panel e, raw, M1b p 0.039; the airgun-suppressed residual does not keep it, p_B 0.30); lag 1,245 s inside the predicted 809-1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests (chance level: a smallest p this low arises about half the time). Arc locus 33.6°S 94.5°E, implied impact 00:31:41; eta for P_D(both) = 0.5 is 0.069 = 330x the F-35A anchor 2.1e-4. BEARING: the locus lies at 269.0° from H01W (1,812 km) and 144.8° from H08S (3,657 km), WGS84; Kadri gives 57° (text p. 9) or 306.18° (Table 1 caption) for the 00:52 signal; neither is within 37° of the locus bearing; nearest Table 1 row 00:52:50 at 274.84° (46 s later). Density: run J impact latitudes (geometry/sample.npz, waypoints seeds 1-8 stride 6 + northern supplement, Kish ESS 110k), D0', 0.1° bins, Gaussian smoothing sigma 0.1°. The x-axis is the latitude of the impact for the density and of the arc crossing for the lines. Davey et al. (2016), Fig. 10.3, recovered from the PDF vector paths (BTO + BFO). Loci E, F, G solved from the lag on the module TL-grid arc at c = 1.482 km/s. Locus C of the earlier chart is not legible and is not redrawn. Source posterior run J, 00:19 Held Out (D0'), dive probability p 0.5; PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated; eta swept over four decades, F-35A upper anchor). Module branch hypothesis/hydroacoustics 9b27786; searches results-data/searches_run_j.